MIL-I-46331(Ord)
15 February 1962
SUPERSEDING
FCS-824A
5 September 1957

MILITARY SPECIFICATION

INDICATOR, AZIMUTH RANGE: IP-289/M

1. SCOPE

1.1 This specification covers one type of precision indicator for indicating azimuth and range and will hereinafter be referred to as "indicator". The indicator is used with antiaircraft and guided missile fire control systems listed in 6.1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts Equipment and Tools for Ordnance Materiel, Packaging of
MIL-F-14619	Fire Control Parts, Equipment and Tools; Packaging of
MIL-I-45208	Inspection Requirements; General Specification for
MIL-I-45607	Inspection Equipment, Supply and Maintenance for Ordnance
MIL-A-45610	Amplifier, Video: AM-1076/M
MIL-G-45611	Generator Sweep: O-294/M
MIL-M-45625	Mixer Stage Sweep Control: CV-318/M
MIL-G-45668	Generator, Pulse: O-295/M

FSC 1285

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:35Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

Ordnance Corps

A7620521	Generator
A7677411	Resolver
E7617884	Indicator, Azimuth and Range: IP-289/M
D7622024	Indicator, Azimuth and Range: IP-289/M, Schematic Diagram

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production one indicator, representative of production, shall be made available to the Government to determine if the indicator meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction.- The indicator shall be manufactured in accordance with Drawing E7617884, "Indicator, Azimuth Range: IP-289/M" and drawings pertaining thereto. The requirements of this specification are detailed only to extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence capabilities prior to final acceptance by the Government.

GPO 818731-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:35Z
Check the source to verify that this is the current version before use.

3.4 Ground potential.- The design and construction of the indicator shall insure that all external parts, exclusive of electrical connections, are at ground potential at all times.

3.5 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Electron tubes
(f) Workmanship

3.6 Environmental.

3.6.1 Operating temperatures.- The indicator shall meet all the performance requirements of this specification when thermally stabilized at ambient temperatures of minus 40 degrees and plus 150 degrees Fahrenheit.

3.6.2 Storage temperatures.- The indicator shall show no evidence of physical failure and shall meet all performance and electrical requirements of this specification at the standard ambient temperature (plus 60 to 90 degrees Fahrenheit) after having been thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit.

3.6.3 Vibration.- The indicator shall be subjected to a simple harmonic motion having an amplitude of 0.015-inch (0.03 inch maximum excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second and return to 10 cycles in approximately 1 minute. This motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total of 30 minutes). Subsequent to vibration the indicator shall not be damaged or adversely affected and shall meet all requirements of this specification.

3.7 Electrical.

3.7.1 Circuitry.- The circuitry of the indicator shall be in accordance with the electrical schematic, Drawing D7622024.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:35Z
Check the source to verify that this is the current version before use.

3.7.2 Dielectric strength.- There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength, when the insulated areas of the indicator are subjected to an alternating current (ac) root mean square (rms) potential of 1000 volts plus twice the rated voltage at a frequency of 60 cycles per second for a period of not more than 30 seconds nor less than 5 seconds.

3.7.3 Insulation resistance.- Unless otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground and between isolated circuits, shall have a minimum resistance of 100 megohms with 500 volts direct current (dc) applied.

3.8 Performance.- When set up in accordance with information specified in Section 4, the indicator shall meet the following requirements. (All symbols referred to in the following paragraphs are those shown on the electrical schematic, Drawing D7622024).

3.8.1 Subassemblies.- The subassemblies comprising the indicator shall have met the requirements of their respective individual specifications as listed below:

Subassembly	Military Specification
Amplifier, Video: AM-1076/M	MIL-A-45610
Generator, Sweep: 0.294/M	MIL-G-45611
Mixer Stage, Sweep Control: CV-318/M	MIL-M-45625
Generator, Pulse: 0-295/M	MIL-G-45668

3.8.2 Focus.- The maximum diameter of the focused spot shall be 1/32-inch.

3.8.3 Centering.- The spot shall be centered horizontally and at the lower extremity of the aperture.

3.8.4 Deflection coil action.- With a range gate in at the Generator, Sweep: 0-294/M (Mixer Stage Sweep Control: CV-318/M out of circuit), R2 of the Generator, Sweep: 0-294/M shall be capable of adjusting the vertical sweep equal to the vertical length of the screen and without curvature. This line shall be visible with the ACQ-TRACK switch in either position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:35Z
Check the source to verify that this is the current version before use.

3.8.5 Range, amplitude and linearity.

3.8.5.1 The 5,000-yard range gate shall be capable of adjusting a series of 1-microsecond (u sec) pips (synchronized with the start of the 5000-yard gate) for 31 pips. The length of the range gate at this particular time shall be not greater than 33 u sec at the 50 percent points.

3.8.5.2 The spaces between the pips (except the first two) shall be equal and the center of the aperture shall be between the 16th and 18th pips from the bottom.

3.8.5.3 Range AMPLITUDE adjust, R2, when fully in the clockwise position shall reduce the number of pips to 26 or less.

3.8.5.4 R2 in the full counterclockwise position shall reduce the sweep to at least 50 percent of aperture height.

3.8.6 Vertical unblanking.- No pips shall be visible on the blanked sweep return trace.

3.8.7 Mixer Stage Sweep Control: CV-318/M balance and sensitivity

3.8.7.1 BAL ADJ Control, R17, shall be capable of having the vertical trace bisect the horizontal line.

3.8.7.2 The trace on the indicator shall be within 1/8-inch of center, when AZ ADJ control, R2, is fully clockwise (resolver scale set to zero).

3.8.7.3 The trace shall sweep horizontally across the tube and out of the aperture to the right, swell, sweep back to the left, swell and return to center, when the resolver is rotated clockwise (sweep started at center of aperture).

3.8.7.4 Rotation of the resolver 12.5 degrees either side of zero (relative to resolver scale) shall drive the sweep beyond the aperture limits.

3.8.7.5 Rotation of the resolver 17.5 degrees either side of zero shall not drive the sweep beyond the aperture, when R2 is fully counterclockwise.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:35Z
Check the source to verify that this is the current version before use.

3.8.7.6 Deviation of the trace from a straight line shall not exceed 3/32 inch, when the sweep is at the left-or right edge of the aperture (R2 reset).

3.8.8 Horizontal unblanking.- The trace shall sweep only from left to right, when the resolver is turned clockwise through one revolution.

3.8.9 Generator, Pulse: 0-295/M.- The width of the azimuth mark shall be 750 u sec, when reolver is at zero.

3.9 Identification marking.- Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 General provisions.- The quality assurance provisions of referenced documents to the extent specified herein, together with provisions of this section, are applicable to inspection performed by the contractor and for Government verification.

4.1.1 The contractor shall provide an inspection system in accordance with Specification MIL-I-45208.

4.2 Production model.- Unless otherwise specified in the contract, one indicator shall be selected by the Government inspector as soon as practicable after the start of production and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.2.1 Production model failure.- Should the production model fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.3 Lot formation Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.3.1 Defective units and lots shall be processed as specified in Standard MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:35Z
Check the source to verify that this is the current version before use.

4.4 Inspection of components.- All positional and functional characteristics, shown on the drawings contained in the contract, which indicate precision requirements and have a bearing on functional accuracy and performance of the indicator shall be inspected in accordance with the written inspection system required by Specification MIL-I-45208. These characteristics are defined as drawing requirements affecting concentricity, squareness, parallelism, micro-finishes 64 RMS or better, threads, identification marking, and performance requirements specified in notes on the drawings and in separate performance specifications referenced on the drawings.

4.5 Material.- Control for requirements of material shall be in accordance with Specification MIL-I-45208.

4.6 Inspection equipment.- Supply, calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL-I-45208. Provisions of Specification MIL-I-45607 shall apply to test equipment provided by the Government or provided by the contractor and utilized in the Government approved inspection system. The Government reserves the right to use all approved test equipment, both Government furnished and contractor acquired for verification inspection.

4.6.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance, except as otherwise specified. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.6.1.1 Test equipment.- The following test equipment accuracy requirements are applicable to the items listed below.

Item No.	Description
1.	Generator shall deliver a 6-volt, 5000 yard positive gate pulse at a 1000 cycles per second repetition rate.
2.	Resolver, Drawing A7677411, 28 volts, 4 kilocycles sine and cosine computer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:35Z
Check the source to verify that this is the current version before use.

Item No. (cont'd)	Description
3.	Generator, Drawing A7620521, 4 kilocycles, to deliver 70 volts and 28 volts sine and cosine components.
4.	Generator, to deliver one (1) microsecond marker pips, synchronized with the start of the 5000 yard gate.
5.	Voltmeter, to measure 6.3 volts, 400 cycles per second. Accuracy of plus or minus 1 percent.
6.	Oscilloscope used to display the input and output pulses shall have sufficient rise time accuracies to display these pulses.
7.	Power supply, plus 450 volts, 0.040 ampere.
8.	Power supply, plus 320 volts, 0.010 ampere.
9.	Power supply, plus 250 volts, 0.100 ampere.
10.	Power supply, plus 150 volts, 0.030 ampere.
11.	Power supply, plus 5500 volts, 0.001 ampere.
12.	Power supply, minus 250 volts, 0.008 ampere.
13.	Power supply, 120 volts ac, 400 cycles per second.
14.	Switch, S2, 5 pole, single throw.
15.	Switch, S3, single pole, single throw.

4.7 Tests.

4.7.1 Classification of defects.- The classification of defects in Tables I, and II shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to inspect for any or all applicable requirements and to reject individual non-conforming items.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:35Z
Check the source to verify that this is the current version before use.

4.7.1.1 In addition to the Standard Measuring Equipment (SME) listed in Table I, items 2 and 3 of paragraph 4.6.1.1, covered by A7677411 and A7620521, respectively, are required for tests as shown in schematic Figure I.

4.7.1.2 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%		Req't Par.	QA. Par.	Test Method
101.	Circuitry	3.7.1	4.8.4.1	Standard Measuring Equipment (SME)
102.	Dielectric strength	3.7.2	4.8.4.2	SME
103.	Insulation resistance	3.7.3	4.8.4.3	SME
104.	Focus	3.8.2	4.9.1	SME
105.	Centering	3.8.3	4.9.2	SME
106.	Deflection coil action	3.8.4	4.9.3	SME
107.	Range, amplitude and linearity	3.8.5	4.9.4	SME
108.	Vertical unblanking	3.8.6	4.9.5	SME
109.	Azimuth sweep generator, balance and sensitivity	3.8.7	4.9.6	SME
110.	Horizontal unblanking	3.8.8	4.9.7	SME
111.	Mark generator test	3.8.9	4.9.8	SME
112.	Identification marking	3.9	4.10	Visual

MINOR: NONE DEFINED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:35Z
Check the source to verify that this is the current version before use.

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 4.0%	Req't Par.	QA. Par.	Test Method
101. Operating temperatures	3.6.1	4.8.1	SME
102. Storage temperatures	3.6.2	4.8.2	SME
103. Vibration	3.6.3	4.8.3	Device, Special Testing (DST)-Visual-Tactile

MINOR: NONE DEFINED

4.8 Environmental.- The test equipment used in conducting temperature tests shall be such that the units under test shall not exceed fifty percent of the internal volume of the test chamber, and no portion shall be closer than six inches from any surface of the chamber.

4.8.1 Operating temperatures.- The units being tested shall be installed in the test chamber in a manner similar to its normal operating position. The test shall then be conducted in accordance with Procedure I of Specification MIL-F-13926 to the conditions specified in 3.6.1. Upon completion of the exposures specified in 3.6.1, the unit under test shall be inspected as depicted herein to determine conformance with the requirements specified.

4.8.2 Storage temperatures.- The indicator to be tested shall be installed in the test chamber in a manner similar to its normal operating position. The test shall then be conducted in accordance with Procedure II of Specification MIL-F-13926 to the conditions specified in 3.6.2. Upon completion of the exposures specified in 3.6.2, the unit under test shall show no evidence of damage by a visual and tactile examination. The unit then shall meet the requirements specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:35Z
Check the source to verify that this is the current version before use.

4.8.3 Vibration.- The indicator shall be clamped on the vibration table in its normal mounting or operating position. The equipment shall be vibrated in each of the three (3) mutually perpendicular directions that are respectively parallel to the edges of the equipment at the acceleration, frequency and duration specified in 3.6.3. At the conclusion of the test, the indicator shall be visually inspected for physical damage, then subjected to the tests specified in 3.6.3.

4.8.4 Electrical.

4.8.4.1 Circuitry.- A continuity check of the indicator circuit shall be made using standard measuring equipment to determine compliance with the requirements specified.

4.8.4.2 Dielectric strength.- The indicator shall be tested in accordance with Method 301 of Standard MIL-STD-202, and shall meet the requirements specified.

4.8.4.3 Insulation resistance.- Immediately after satisfactorily withstanding the test for dielectric strength, the indicator shall be given the insulation resistance test in accordance with Method 302, test condition B of Standard MIL-STD-202, and shall meet the requirements specified.

4.9 Performance.- The tests conducted to determine performance requirements shall be accomplished utilizing the test equipment specified in 4.6.1.1, connected as shown in Figure I, or as described herein.

(a) The test circuit shall be connected as shown in Figure 1, with power off. Switch S2 shall be set in the open position, and switch S3 set in the ground position. The range gate P6 shall be removed from Sweep Generator: O-294/M.

(b) The decentering magnet shall be oriented centrally with respect to its slotted mounting holes.

(c) The deflection coil and focusing magnet shall be concentric within 1/32 inch.

(d) The cathode ray tube neck shall have equal clearance around the rear of the focusing magnet and the front of the deflection coil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:35Z
Check the source to verify that this is the current version before use.

(e) The GAIN control, R9, and INTENSITY control, R5, shall be set at their full counterclockwise position.

(f) Power shall be applied to the indicator as in Figure 1.

(g) With all of the units inserted, the 400 cycle per second filament voltage shall be measured between terminals 14 and 15 of J8, J9, J10 and P3; and between pins 9 and 11 of P3. The voltage measured shall be 6.3 plus or minus 0.3 volts as indicated on voltmeter, item 5.

4.9.1 Focus.- The focused spot shall be as specified in 3.8.2 when the INTENSITY control is increased until a normal spot is visible on the screen.

4.9.2 Centering.- The front adjusting plate of the decentering magnet shall be rotated until the spot is at the bottom of the aperture and horizontally centered as specified in 3.8.3.

4.9.3 Deflection coil action.

4.9.3.1 The range gate P6 shall be reconnected to the Generator, Sweep: 0-294/M and the INTENSITY control R5 adjusted until a vertical line is visible on the screen of the oscilloscope, item 6. This line shall be visible with Acq-Track switch S1 in either position.

4.9.3.2 The AMPLITUDE control R2 of Sweep Generator: 0-294/M shall be adjusted until the vertical sweep shall be as specified in 3.8.4.

4.9.4 Range, amplitude and linearity.

4.9.4.1 A series of one (1) microsecond pips synchronized with the start of the 5000 yard gate shall be applied to the video input J4. The gain shall be raised until the pips are visible on the screen. The 5000 yard gate shall be adjusted until the number of pips and length of the range gate at J6 are as specified in 3.8.5.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:35Z
Check the source to verify that this is the current version before use.

4.9.4.2 With the exception of the first two pips, the spacing between the pips shall be as specified in 3.8.5.2.

4.9.4.3 R2 of the Sweep Generator: O-294/M shall be turned to its full clockwise position. The number of visible pips shall be reduced and shall not exceed the number specified in 3.8.5.3.

4.9.4.3 R2 shall be turned to its full counterclockwise position and the sweep shall be reduced as specified in 3.8.5.4.

4.9.5 Vertical unblanking.- The gain shall be reduced and the intensity increased until a faint trace shall be visible. The gain shall then be increased until the dots begin to effloresce and there shall be no visible pips on the blanked sweep return trace as specified in 3.8.6.

4.9.6 Mixer Stage, Sweep Control: CV-318 balance and sensitivity.

4.9.6.1 The Sweep Control Mixer Stage: CV-318/M shall be plugged in. The balance adjust control R17 shall be adjusted so that the vertical trace is as specified in 3.8.7.1.

4.9.6.2 A 28-volt, 4 kilocycle per second input shall be applied to the resolver, item 2, and switch S2 shall be set to the ON position. The AZ ADJ control, R2, shall be advanced to its full clockwise position and the resolver scale set to zero. The trace on the indicator shall be as specified in 3.8.7.2.

4.9.6.3 The resolver degree knob shall be turned clockwise through 360 degrees and the trace shall sweep as specified in 3.8.7.3.

4.9.6.4 The resolver shall be rotated 12.5 degrees either side of zero to drive the sweep as specified in 3.8.7.4.

4.9.6.5 The AZ ADJ control, R2, shall be set to its full counterclockwise position and the resolver rotated 17.5 degrees either side of zero. The drive of the sweep shall be as specified in 3.8.7.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:35Z
Check the source to verify that this is the current version before use.

4.9.6.6 The trace shall be deflected to within 1/8 inch of the aperture edge when the AZ ADJ control, R2, has set the resolver 15 degrees in either direction off center. Deviation of the trace shall not exceed the limits specified in 3.8.7.6.

4.9.7 Horizontal unblanking.- The resolver shall be set for 15.1 degrees and the AZIMUTH BLANK ADJ, R2, on the Video Amplifier: AM-1076/M adjusted so that the trace is just blanked. The resolver shall then be turned clockwise through one revolution and the trace shall sweep as specified in 3.8.8.

4.9.8 Generator, Pulse: 0-295/M test.- The resolver dial shall be set to zero. The width of the azimuth mark shall be as specified in 3.8.9.

4.10 Identification marking.- The identification marking shall be visually inspected for compliance with Standard MIL-STD-130.

4.11 Quality assurance provisions for packaging, packing and marking.- Quality assurance provisions for packaging, packing and marking shall be in accordance with Specification MIL-P-14232.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- The indicator shall be cleaned, preserved, packaged, packed, and marked in accordance with Specification MIL-F-14619 P7229. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The purpose of the indicator is to provide a B type display for either 30 degrees azimuth and 5000 yard range or 30 degrees azimuth and 8000 yard range depending on the output of the range sweep generator. The indicator is a component of the following assemblies and systems:

ASSEMBLY	SYSTEM
Console, Tracking 7622329 and Console, Tacticle Control 7604182	Aircraft Fire Control System M33D (120MM) 7603783
Console, Tracking 7622329 and Console, Tactical Control 7604175	Antiaircraft Fire Control System M33C (90MM) 7603966

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:35Z
Check the source to verify that this is the current version before use.

ASSEMBLY	SYSTEM
Test Cabinet Assembly 7603980	Electronic Field-Maintenance Shop Set for Antiaircraft Fire Control System M33 Series 7651254
Console, Battery Control OA-723/MSA-7 and Console Radar Control OA-724/MPA-4	Radar Course Directing Central AN/MSW-1
Console, Tracking OA-606/MSQ-3 and Console, Tactical Control OA-607/MSG-3	Guided Missile Fire Control System AN/MSG-3

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable level of preservation, packaging and packing
(c) Applicable stock number
(d) Provisions for production model including date of delivery (see 3.1)

6.3 Definitions.

6.3.1 Standard measuring equipment (SME).- Standard measuring equipment includes the common, hand-type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

6.3.2 Special testing device(DST).- Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

6.3.3 Tactile.- Skillful manipulation or sensing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:35Z
Check the source to verify that this is the current version before use.

6.4 Design information.- Design information regarding specific equipment not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder, or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

Custodian:

Army-Ord

Preparing activity:

Army-Ord

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:35Z
Check the source to verify that this is the current version before use.

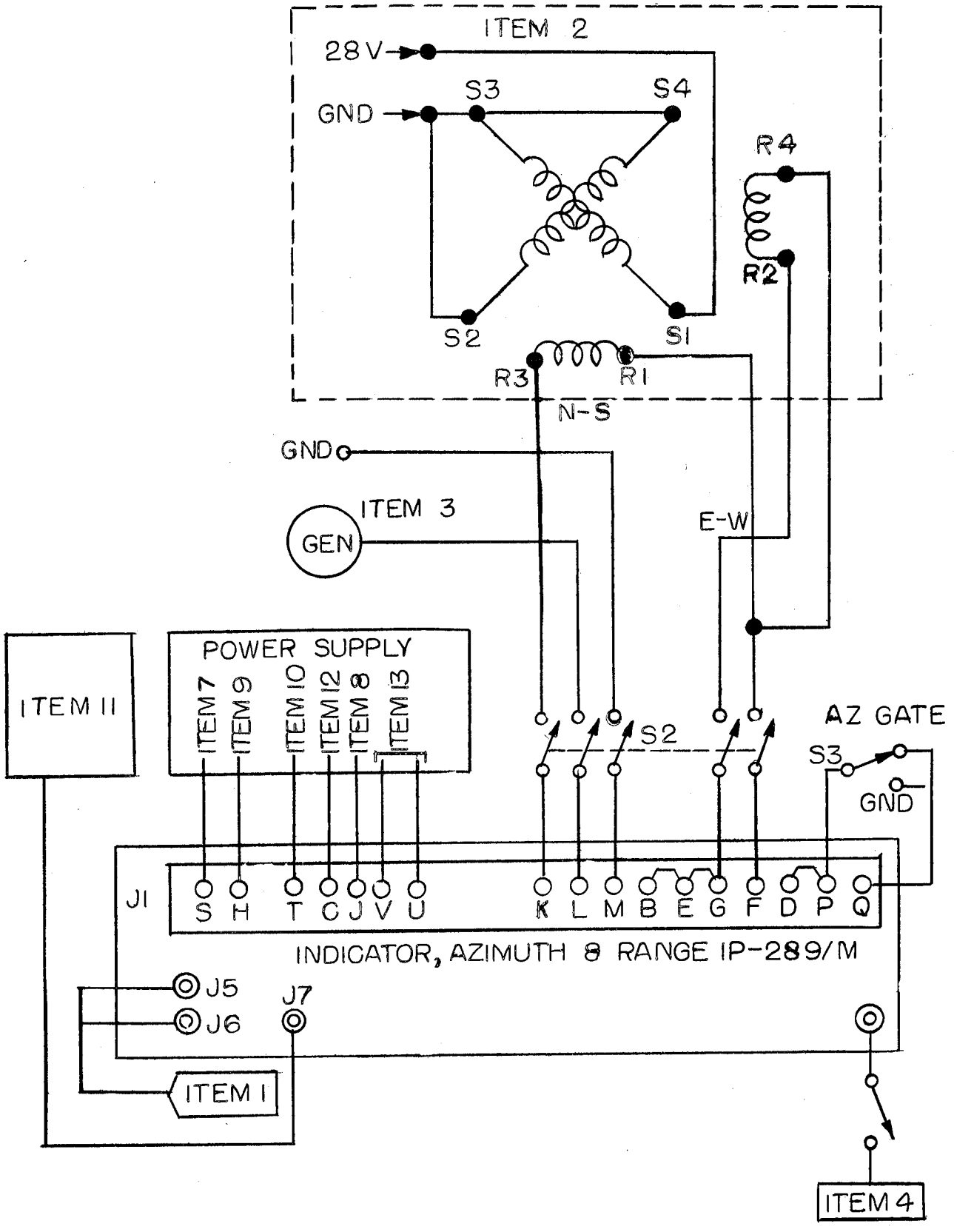


FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:35Z
Check the source to verify that this is the current version before use.